

Name - Divij Mahajan

ID - 23BCS12775

Class - 23BCS KRG-1-A

Find sum of $f(A_i, A_j)$ where $f(x, y)$ is no. of diff. bits. Return ans by doing $10^9 + 7$ modulo.

§. For each bit i $\rightarrow$ $1 \rightarrow 31$ $(2^0) \quad (2^{30})$

• For full array.

• Count active bit (1) and inactive bit (0)

mod $\rightarrow 10^9 + 7$ • multiply active bit count with inactive bit count also do mod.

• Add this to $\{ans = (ans + product) \% mod$

• Finally return $(ans \times 2) \% mod$.

Code:

```
int countDiffBits(vector<int> &arr) {  
    long long ans = 0, active = 0;  
    int n = arr.size(), mod = 1000000007;  
    for (int i = 0; i < 31; i++) {  
        if (active == 0;  
            for (int j = 0; j < n; j++) {  
                if (arr[j] & (1 << i))  
                    active++;  
            }  
        }  
    }  
}
```


}

long long prod = active * (n - active);

prod = prod % mod;

ans = (ans + prod) % mod;

}

return (ans * 2) % mod;

}

Test case

1) 1, 3, 5

$\downarrow$ $\downarrow$ $\downarrow$
001 011 101

active inactive prod

1st $\rightarrow$ bit 3 0 0

2nd $\rightarrow$ bit 1 2 2

3rd $\rightarrow$ bit 1 2 2

total $\rightarrow$ 4

As we will consider $f(2, 3)$ & $f(3, 2)$ also

so total * 2 $\rightarrow$ 8

OUTPUT = 8

$$2.) \quad \begin{array}{c} 2, 3 \\ \downarrow \quad \downarrow \\ 10 \quad 11 \end{array}$$

	Active	Inactive	Read
<u>1st</u>	1	1	1
<u>2nd</u>	1	0	0
			<u>2</u>

$$\text{ans} = 1 \times 2 \Rightarrow 2$$

OUTPUT $\rightarrow 2$

$$\begin{aligned} \text{Time complexity} &= 32 \times n \\ &= O(n) \end{aligned}$$

$$\text{space complexity} = O(1)$$